JS Weekend Question Bank

Please mention the Time and Space Complexity for each of the questions

Q1. Solid Right Triangle

Print a right triangle of `*` with `n` rows.

Input

- `n = 4`

Output

```
*
**
***
****
```

Q2. Inverted Right Triangle

Print an inverted right triangle of `*` with `n` rows.

Input

- `n = 5`

Output

```
*****
****
***
**
*
```

Q3. Centered Pyramid (1, 3, 5, 7)

Print a centered pyramid of `*` with `n` rows.

Input

- `n = 4`

Output

```
   *
  ***
 *****
*******
```

Q4. Diamond Pattern (1, 2, 3, 4, 5)

Print a diamond of `*` with odd height `n`.

Input

- `n = 4`

Output

```
   *
  * *
 * * *
* * * *
```

Q5. Row and Column Sums

Given a 2D matrix, print sum of each row on one line, then sum of each column on the next line.

Input

```
matrix = [
  [1, 2, 3],
  [4, 5, 6],
  [7, 8, 9]
]
```

Output

```
Row sums: 6 15 24
Column sums: 12 15 18
```

Q6. Transpose Matrix

Given an `m x n` matrix, print its transpose `n x m`.

Input

```
matrix = [
  [1, 2, 3],
  [4, 5, 6]
]
```

Output

```
[
  [1, 4],
  [2, 5],
  [3, 6]
]
```

Q7. Flatten One-Level Nested Array

Flatten an array that may contain numbers or one-level nested arrays.

Input

```
arr = [1, [2, 3], 4, [5]]
```

Output

```
[1, 2, 3, 4, 5]
```

Q8. Longest Word in 2D Char Grid

Each row is a sequence of characters with `'#'` as separator. Find the longest continuous word.

Input

```
grid = [
  ['c','a','t','#','d','o','g'],
  ['a','p','p','l','e','#','#']
]
```

Output

```
apple
```

Q9. What is the value of count when the code is done executing (this is a predict the output type question, solve it using pen and paper).

Given the function:

```
function mystery(n) {
  let count = 0;
  for (let i = 0; i < n; i++) {
    for (let j = 0; j < n; j++) {
      count++;
    }
  }
  return count;
}
```

Input

```
n = 5
```

Q10. What is the value of count when the code is done executing (this is a predict the output type question, solve it using pen and paper).

Given:

```
function fun(n) {
  let count = 0;
  while (n > 0) {
    n = Math.floor(n / 2);
    count++;
  }
  return count;
}
```

Input

```
n = 20
```

Q11. Iterative Binary Search

Implement binary search on a sorted array; return index of target or `-1`.

Input

```
arr = [1, 3, 5, 7, 9, 11]
target = 7
```

Output

```
3
```

Q12. First Occurrence in Sorted Array

Return the index of the **first** occurrence of `target` in a sorted array with duplicates.

Input

```
arr = [1, 2, 2, 2, 3, 4]
target = 2
```

Output

```
1
```

Q13. Character Frequency

Count the frequency of each character in a string using `Map`. Print character and its count.

Input

```
str = "banana"
```

Output (order can vary)

```
b 1
a 3
n 2
```

Q14. Remove Duplicates with Set

Return a new array with duplicates removed.

Input

```
arr = [1, 2, 2, 3, 1, 4, 4]
```

Output

```
[1, 2, 3, 4]
```

Q15. Sum of Array Recursively

Write a recursive function to sum all elements of an array.

Input

```
arr = [2, 4, 6, 8]
```

Output

```
20
```

Q16. Recursive String Reverse

Reverse a string using recursion (no built-in `.reverse()`).

Input

```
str = "hello"
```

Output

```
olleh
```

Q17. Bubble Sort

Sort an array in ascending order using bubble sort.

Input

```
arr = [5, 1, 4, 2, 8]
```

Output

```
[1, 2, 4, 5, 8]
```

Q18. Selection Sort

Sort an array in ascending order using selection sort.

Input

```
arr = [64, 25, 12, 22, 11]
```

Output

```
[11, 12, 22, 25, 64]
```

Q19. Insertion Sort

Sort an array in ascending order using insertion sort.

Input

```
arr = [9, 5, 1, 4, 3]
```

Output

```
[1, 3, 4, 5, 9]
```

Q20. Merge Two Sorted Arrays

Merge two sorted arrays into one sorted array.

Input

```
left = [1, 3, 5]
right = [2, 4, 6, 8]
```

Output

```
[1, 2, 3, 4, 5, 6, 8]
```

Q21. Merge Sort on Array

Sort an array using merge sort.

Input

```
arr = [7, 2, 6, 3, 9, 1]
```

Output

```
[1, 2, 3, 6, 7, 9]
```

Q22. Quick Sort

Write the Quick Sort algorithm for both linear and constant space complexity